Post Correspondence Problem

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Reducibility So Far

- To prove that E_{TM} is undecidable, we need to show that if E_{TM} is decidable, then A_{TM} is decidable.
  1. Assume that E_{TM} is decidable and obtain its decider R
  2. Construct TM S with the help of R to decide A_{TM}
     - This TM S takes $\langle M, w \rangle$ (an instance of A_{TM}) and construct a new TM M'
     - $L(M') \neq \emptyset$ if M accepts w
     - $L(M') = \emptyset$ if M does not accepts w (either rejects or loops indefinitely)
  3. Prove that TM S is a decider for A_{TM}
  4. Conclude that A_{TM} is decidable — contradiction
  5. Therefore, E_{TM} is undecidable

Mapping Reducibility So Far

- To prove that EQ_{TM} is undecidable, we need to show that A_{TM} is mapping reducible to EQ_{TM}
  1. Create a computable function f in the form of a TM F
     - This TM F takes $\langle M, w \rangle$ (an instance of A_{TM}) and construct two TMs M_1 and M_2
     - $L(M_1) = L(M_2)$ if M accepts w
     - $L(M_1) \neq L(M_2)$ if M does not accepts w (either rejects or loops indefinitely)
     - Output $\langle M_1, M_2 \rangle$ (an instance of EQ_{TM})
  2. Prove that $\langle M, w \rangle \in A_{\mathsf{TM}} \leftrightarrow \langle M_1, M_2 \rangle \in EQ_{\mathsf{TM}}$
  3. Since A_{TM} is undecidable, EQ_{TM} is undecidable

Reduction via Computation History

- To prove that E_{LBA} is undecidable, we need to show that if E_{LBA} is decidable, then A_{TM} is decidable.
  1. Assume that E_{LAB} is decidable and obtain its decider R
  2. Construct TM S with the help of R to decide A_{TM}
     - This TM S take $\langle M, w \rangle$ (an instance of A_{TM}) an construct a new LBA M'
     - This LBA M' only accepts the *accepting computation history for M on input w*
     - $L(M') \neq \emptyset$ if M accepts w
     - $L(M') = \emptyset$ if M does not accept w (either rejects or loops indefinitely)
  3. Prove that S is a decider for A_{TM}
  4. Conclude that A_{TM} is decidable — contradiction
  5. Therefore, E_{LBA} is undecidable

Common Theme

- All three methods so far share a common process
  - Take $\langle M, w \rangle$ (an instance of A_{TM})
  - Construct a new TM, TMs, or LBA
  - Behaviors of new TM, TMs, or LBA depend on whether M accepts w
- Construct TM or TMs from TM M and string w are quite straightforward
  - We can simply run M on input w
- Construct LBA from TM M and string w is not quite straightforward
  - We cannot simply run M on input w
  - Use computation history (sequence of configurations)

Post Correspondence Problem (PCP)

- A **Post Correspondence Problem** (PCP) is a problem concerning string manipulation.
- Consider a domino

$$\left[\frac{\texttt{a}}{\texttt{ab}}\right]$$

There are two non-empty strings in each domino, one on the top and one on the bottom.

- Given a set of dominoes

$$\left\{\left[\frac{\texttt{b}}{\texttt{ca}}\right], \left[\frac{\texttt{a}}{\texttt{ab}}\right], \left[\frac{\texttt{ca}}{\texttt{a}}\right], \left[\frac{\texttt{abc}}{\texttt{c}}\right]\right\}$$

Is it possible to make a horizontal line of one or more dominoes, with duplicates allowed, so that the string obtained by reading across the top halves **matches** the one obtained by reading across the bottom?

Post Correspondence Problem (PCP)

- For this set of dominoes:

$$\left\{ \left[\frac{\texttt{b}}{\texttt{ca}}\right], \left[\frac{\texttt{a}}{\texttt{ab}}\right], \left[\frac{\texttt{ca}}{\texttt{a}}\right], \left[\frac{\texttt{abc}}{\texttt{c}}\right] \right\}$$

  There is a **match** as shown below:

$$\left[\frac{\texttt{a}}{\texttt{ab}}\right]\left[\frac{\texttt{b}}{\texttt{ca}}\right]\left[\frac{\texttt{ca}}{\texttt{a}}\right]\left[\frac{\texttt{a}}{\texttt{ab}}\right]\left[\frac{\texttt{abc}}{\texttt{c}}\right]$$

- The following set of dominoes does not have a match:

$$\left\{ \left[\frac{\texttt{abc}}{\texttt{ab}}\right], \left[\frac{\texttt{ca}}{\texttt{a}}\right], \left[\frac{\texttt{acc}}{\texttt{ba}}\right] \right\}$$

  - For obvious reason, each top string is longer than its bottom string.

Post Correspondence Problem (PCP)

- An instance P of PCP is a collection of dominoes

$$P = \left\{ \left[\frac{t_1}{b_1}\right], \left[\frac{t_2}{b_2}\right], \ldots, \left[\frac{t_k}{b_k}\right], \right\}$$

- A **match** is a sequence $i_1, i_2, \ldots, i_l$ such that

$$t_{i_1} t_{i_2} \ldots t_{i_l} = b_{i_1} b_{i_2} \ldots b_{i_l}$$

- As usual, we define PCP as a set

$$PCP = \{\langle P \rangle \mid P \text{ is an instance of the Post Correspondence Problem with a match}\}.$$

- Is PCP decidable?
  - Given an instance P (a set of dominoes) of *PCP*, can we decide whether $P \in PCP$ or not?
    - $P \in PCP$ if P has a match
    - $P \notin PCP$ if P does not have a match

Post Correspondence Problem (PCP)

- To prove that PCP is undecidable, we use the same method as what we did for other undecidable languages
  1. Assume that *PCP* is decidable and obtain its decider R
  2. Construct TM S with the help of R to decide A_{TM}
     - Take $\langle M, w \rangle$ (an instance of A_{TM}) and construct P **an instance of PCP which is a set of dominos**
     - P has a match if M accepts w
     - P does not have a match if M does not accept w
  3. Prove that S is a decider for A_{TM}
  4. Conclude that A_{TM} is decidable — contradiction
  5. Therefore, *PCP* is undecidable
- Note that we have to construct a set of **dominoes** from TM M and string w
  - This process is long but straightforward
  - We also need to look at another closely related problem

Modified Post Correspondence Problem (MPCP)

- A **Modified Post Correspondence Problem** (MPCP) looks exactly like PCP but the match is required to start with the first domino.
- An instance P of MPCP is a collection of dominoes

$$P = \left\{ \left[\frac{t_1}{b_1}\right], \left[\frac{t_2}{b_2}\right], \ldots, \left[\frac{t_k}{b_k}\right], \right\}$$

- A **match** is a sequence 1,i_2, i_3, $\ldots$, i_l such that

$$t_1 t_{i_2} \ldots t_{i_l} = b_1 b_{i_2} \ldots b_{i_l}$$

- As usual, we define MPCP as a language

$MPCP = \{\langle P \rangle \mid P$ is an instance of the Post Correspondence Problem with a match that starts with he first domino$\}$.

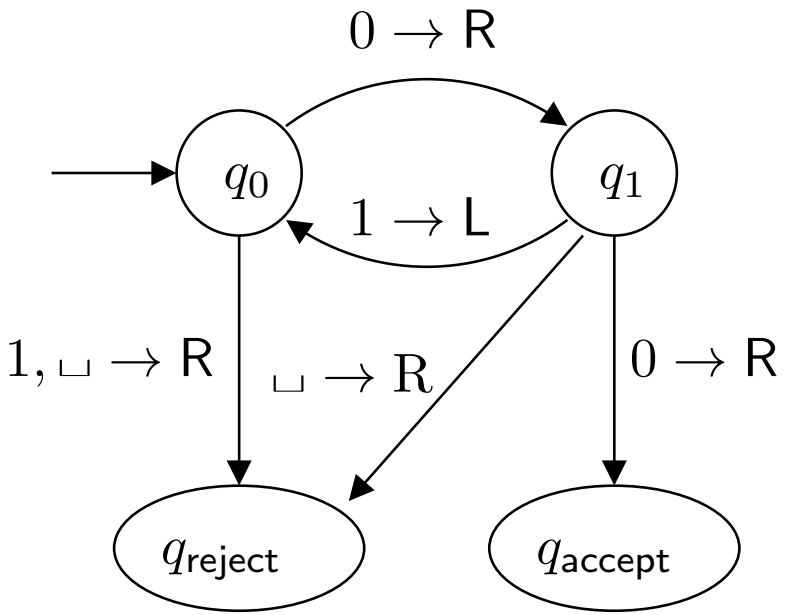
- We are going to construct P' (an instance of *MPCP*) from TM M and string w such that P' has a match if an only if M accepts w

Constructing a Set of Dominoes

- For simplicity, we are going to pick a specific TM M and a string w
- Construct a set of dominoes P' from M and w
- Verify that P' has a match if and only if M accepts w

Converting a TM and a string to a set of dominoes

- Consider the following TM M:



- The above machine:
  - *accept* 00:

$$q_0 00 \to 0 q_1 0 \to 00 q_{\text{accept}}$$

  - *reject* 1:

$$q_0 1 \to 1 q_{\text{reject}}$$

  - *loops indefinitely on* 01:

$$q_0 01 \to 0 q_1 1 \to q_0 01 \to 0_q 11 \to \ldots$$

Computation History

- Recall that M accepts w if and only if **accepting computation history for M on input w exists**
- Since M (from previous slide) accepts 00, the accepting computation history for M on input 00 exists

$$q_0 00 \rightarrow 0 q_1 0 \rightarrow 00 q_{\text{accept}}$$

- We can turn it into a string as

$$\#q_0 00 \# 0 q_1 0 \# 00 q_{\text{accept}} \#$$

- This will be our match
  - If the accepting computation history for M on input w exist, there is a match
    - And the match looks exactly like the computation history
  - If the accepting computation history for M on input w does not exist, there is no match

Formal Definition of TM M

- The formal definition of the previous TM M is as follows:
  - $Q = \{q_0, q_1\}$
  - $\Sigma = \{0, 1\}$
  - $\Gamma = \{0, 1, \sqcup\}$
  - δ are as follows:
    - $\delta(q_0, 0) = (q_1, 0, \mathsf{R})$
    - $\delta(q_0, 1) = (q_{\text{reject}}, 1, \mathsf{R})$
    - $\delta(q_0, \sqcup) = (q_{\text{reject}}, \sqcup, \mathsf{R})$
    - $\delta(q_1, 0) = (q_{\text{accept}}, 0, \mathsf{R})$
    - $\delta(q_1, 1) = (q_0, 1, \mathsf{L})$
    - $\delta(q_1, \sqcup) = (q_{\text{reject}}, \sqcup, \mathsf{R})$
  - q_0 is the start state
  - q_{accept}
  - q_{reject}

Converting a TM and a string to a set of dominoes

- For the TM M (from earlier slide) and the string 00, let's construct a set of dominoes (an instance of MPCP):
  - First, create the domino where the starting configuration is at the bottom part:

$$\left[\frac{\#}{\#q_0 00\#}\right]$$

    This is the first domino, **a match must start with this domino**.
- This is what we have so far:

$$\left\{\left[\frac{\#}{\#q_0 00\#}\right]\right\}$$

Converting a TM and a string to a set of dominoes

- Consider a configuration

$$ucqav$$

where u and v are strings, a and c are symbols in Γ, and q is a state

- Suppose M has a transition as follows:

$$\delta(q, a) = (r, b, \mathsf{R})$$

- The above configuration yields

$$ucbrv$$

- If we put one of top of the other:

$\frac{ucqav}{ucbrv}$ the part that change from top to bottom is $\frac{qa}{br}$

which will be our dominoes

Converting a TM and a string to a set of dominoes

- For every $a, b \in \Gamma$ and every $q, r \in Q$ where $q \neq q_{\text{reject}}$

$$\text{if } \delta(q, a) = (r, b, \text{R}), \text{ put } \left[\frac{qa}{br}\right] \text{ into the set}$$

- There are five transitions that move the tape head to the right direction:
  - $\delta(q_0, 0) = (q_1, 0, \text{R})$ gives us $\left[\frac{q_0 0}{0 q_1}\right]$
  - $\delta(q_0, 1) = (q_{\text{reject}}, 1, \text{R})$ gives us $\left[\frac{q_0 1}{1 q_{\text{reject}}}\right]$
  - $\delta(q_0, \sqcup) = (q_{\text{reject}}, \sqcup, \text{R})$ gives us $\left[\frac{q_0 \sqcup}{\sqcup q_{\text{reject}}}\right]$
  - $\delta(q_1, 0) = (q_{\text{accept}}, 0, \text{R})$ gives us $\left[\frac{q_1 0}{0 q_{\text{accept}}}\right]$
  - $\delta(q_1, \sqcup) = (q_{\text{reject}}, \sqcup, \text{R})$ gives us $\left[\frac{q_1 \sqcup}{\sqcup q_{\text{reject}}}\right]$

Converting a TM and a string to a set of dominoes

- This is what we have so far:

$$\left\{ \left[\frac{\#}{\#q_0 00\#}\right], \right.$$
$$\left. \left[\frac{q_0 0}{0 q_1}\right], \left[\frac{q_0 1}{1 q_{\text{reject}}}\right], \left[\frac{q_1 0}{0 q_{\text{accept}}}\right], \left[\frac{q_0 \sqcup}{\sqcup q_{\text{reject}}}\right], \left[\frac{q_1 \sqcup}{\sqcup q_{\text{reject}}}\right] \right\}$$

Converting a TM and a string to a set of dominoes

- Consider a configuration

$$ucqav$$

where u and v are strings, c and a are symbols in Γ, and q is a state

- Suppose M has a transition as follows:

$$\delta(q, a) = (r, b, \mathsf{L})$$

- The above configuration yields

$$urcbv$$

- If we put one of top of the other:

$\frac{ucqav}{urcbv}$ the part that change from top to bottom is $\frac{cqa}{rcb}$

which will be our dominoes

Converting a TM and a string to a set of dominoes

- For every $a, b, c \in \Gamma$ and every $q, r \in Q$ where $q \neq q_{\text{reject}}$

$$\text{if } \delta(q, a) = (r, b, \mathsf{L}), \text{ put } \left[\frac{cqa}{rcb}\right] \text{ into the set}$$

- There is only one transition that move the tape head to the left direction:
  - $\delta(q_1, 1) = (q_0, 1, \mathsf{L})$ gives us the following dominoes:

$$\left[\frac{0q_11}{q_001}\right], \left[\frac{1q_11}{q_011}\right], \text{ and } \left[\frac{\sqcup q_11}{q_0\sqcup 1}\right]$$

Converting a TM and a string to a set of dominoes

- This is what we have so far:

$$\left\{ \left[\frac{\#}{\#q_0 00\#}\right], \right.$$
$$\left[\frac{q_0 0}{0 q_1}\right], \left[\frac{q_0 1}{1 q_{\text{reject}}}\right], \left[\frac{q_1 0}{0 q_{\text{accept}}}\right], \left[\frac{q_0 \sqcup}{\sqcup q_{\text{reject}}}\right], \left[\frac{q_1 \sqcup}{\sqcup q_{\text{reject}}}\right],$$
$$\left.\left[\frac{0 q_1 1}{q_0 01}\right], \left[\frac{1 q_1 1}{q_0 11}\right], \left[\frac{\sqcup q_1 1}{q_0 \sqcup 1}\right] \right\}$$

Converting a TM and a string to a set of dominoes

- For every $a \in \Gamma$, put $\left[\frac{a}{a}\right]$ into the set
  - Our Γ is $\{0, 1, \sqcup\}$, this gives us the following dominoes:

$$\left[\frac{0}{0}\right], \left[\frac{1}{1}\right], \text{ and } \left[\frac{\sqcup}{\sqcup}\right]$$

- Next, put $\left[\frac{\#}{\#}\right]$ and $\left[\frac{\#}{\sqcup\#}\right]$ into the set

Converting a TM and a string to a set of dominoes

- This is what we have so far:

$$\left\{ \left[\frac{\#}{\#q_0 00\#}\right], \right.$$
$$\left[\frac{q_0 0}{0 q_1}\right], \left[\frac{q_0 1}{1 q_{\text{reject}}}\right], \left[\frac{q_1 0}{0 q_{\text{accept}}}\right], \left[\frac{q_0 \sqcup}{\sqcup q_{\text{reject}}}\right], \left[\frac{q_1 \sqcup}{\sqcup q_{\text{reject}}}\right],$$
$$\left[\frac{0 q_1 1}{q_0 01}\right], \left[\frac{1 q_1 1}{q_0 11}\right], \left[\frac{\sqcup q_1 1}{q_0 \sqcup 1}\right],$$
$$\left.\left[\frac{0}{0}\right], \left[\frac{1}{1}\right], \left[\frac{\sqcup}{\sqcup}\right], \left[\frac{\#}{\#}\right], \left[\frac{\#}{\sqcup\#}\right] \right\}$$

Converting a TM and a string to a set of dominoes

- For every $a \in \Gamma$, put $\left[\frac{aq_{\text{accept}}}{q_{\text{accept}}}\right]$ and $\left[\frac{q_{\text{accept}}a}{q_{\text{accept}}}\right]$ into the set.
  - This part gives us the following dominoes:

$$\left[\frac{0q_{\text{accept}}}{q_{\text{accept}}}\right], \left[\frac{1q_{\text{accept}}}{q_{\text{accept}}}\right], \left[\frac{\sqcup q_{\text{accept}}}{q_{\text{accept}}}\right], \left[\frac{q_{\text{accept}}0}{q_{\text{accept}}}\right], \left[\frac{q_{\text{accept}}1}{q_{\text{accept}}}\right], \left[\frac{q_{\text{accept}}\sqcup}{q_{\text{accept}}}\right]$$

- Lastly, simply add the domino $\left[\frac{q_{\text{accept}}\#\#}{\#}\right]$

Converting a TM and a string to a set of dominoes

- For the TM M and the string 00, we get this set of dominoes:

$$\left\{ \left[\frac{\#}{\#q_0 00\#}\right], \right.$$
$$\left[\frac{q_0 0}{0 q_1}\right], \left[\frac{q_0 1}{1 q_{\text{reject}}}\right], \left[\frac{q_1 0}{0 q_{\text{accept}}}\right], \left[\frac{q_0 \sqcup}{\sqcup q_{\text{reject}}}\right], \left[\frac{q_1 \sqcup}{\sqcup q_{\text{reject}}}\right],$$
$$\left[\frac{0 q_1 1}{q_0 01}\right], \left[\frac{1 q_1 1}{q_0 11}\right], \left[\frac{\sqcup q_1 1}{q_0 \sqcup 1}\right],$$
$$\left[\frac{0}{0}\right], \left[\frac{1}{1}\right], \left[\frac{\sqcup}{\sqcup}\right], \left[\frac{\#}{\#}\right], \left[\frac{\#}{\sqcup\#}\right],$$
$$\left[\frac{0 q_{\text{accept}}}{q_{\text{accept}}}\right], \left[\frac{1 q_{\text{accept}}}{q_{\text{accept}}}\right], \left[\frac{\sqcup q_{\text{accept}}}{q_{\text{accept}}}\right],$$
$$\left[\frac{q_{\text{accept}} 0}{q_{\text{accept}}}\right], \left[\frac{q_{\text{accept}} 1}{q_{\text{accept}}}\right], \left[\frac{q_{\text{accept}} \sqcup}{q_{\text{accept}}}\right], \left[\frac{q_{\text{accept}} \#\#}{\#}\right] \right\}$$

The set of dominoes

- Let's focus on the set of dominoes that we just constructed:

$$\left\{ \left[\frac{\#}{\#q_0 00\#}\right], \left[\frac{q_0 0}{0 q_1}\right], \left[\frac{q_0 1}{1 q_{\text{reject}}}\right], \left[\frac{q_1 0}{0 q_{\text{accept}}}\right], \left[\frac{q_0 \sqcup}{\sqcup q_{\text{reject}}}\right], \left[\frac{q_1 \sqcup}{\sqcup q_{\text{reject}}}\right], \left[\frac{0 q_1 1}{q_0 01}\right], \right.$$
$$\left[\frac{1 q_1 1}{q_0 11}\right], \left[\frac{\sqcup q_1 1}{q_0 \sqcup 1}\right], \left[\frac{0}{0}\right], \left[\frac{1}{1}\right], \left[\frac{\sqcup}{\sqcup}\right], \left[\frac{\#}{\#}\right], \left[\frac{\#}{\sqcup\#}\right], \left[\frac{0 q_{\text{accept}}}{q_{\text{accept}}}\right], \left[\frac{1 q_{\text{accept}}}{q_{\text{accept}}}\right],$$
$$\left.\left[\frac{\sqcup q_{\text{accept}}}{q_{\text{accept}}}\right], \left[\frac{q_{\text{accept}} 0}{q_{\text{accept}}}\right], \left[\frac{q_{\text{accept}} 1}{q_{\text{accept}}}\right], \left[\frac{q_{\text{accept}} \sqcup}{q_{\text{accept}}}\right], \left[\frac{q_{\text{accept}} \#\#}{\#}\right] \right\}$$

- If we consider q_0, q_1, q_{accept} and q_{reject} as symbols (not strings)
  - We can categorize dominoes into three groups based on the length of the top string vs the bottom string

The set of dominoes

- The top string is shorter than the bottom string:

$$\left[\frac{\#}{\#q_0 00\#}\right], \left[\frac{\#}{\sqcup\#}\right]$$

- The top string and the bottom string have equal length:

$$\left[\frac{q_0 0}{0 q_1}\right], \left[\frac{q_0 1}{1 q_{\mathsf{r}}}\right], \left[\frac{q_1 0}{0 q_{\mathsf{a}}}\right], \left[\frac{q_0 \sqcup}{\sqcup q_{\mathsf{r}}}\right], \left[\frac{q_1 \sqcup}{\sqcup q_{\mathsf{r}}}\right],$$
$$\left[\frac{0 q_1 1}{q_0 01}\right], \left[\frac{1 q_1 1}{q_0 11}\right], \left[\frac{\sqcup q_1 1}{q_0 \sqcup 1}\right], \left[\frac{0}{0}\right], \left[\frac{1}{1}\right], \left[\frac{\sqcup}{\sqcup}\right], \left[\frac{\#}{\#}\right]$$

- The top string is longer than the bottom string:

$$\left[\frac{0 q_{\mathsf{a}}}{q_{\mathsf{a}}}\right], \left[\frac{1 q_{\mathsf{a}}}{q_{\mathsf{a}}}\right], \left[\frac{\sqcup q_{\mathsf{a}}}{q_{\mathsf{a}}}\right], \left[\frac{q_{\mathsf{a}} 0}{q_{\mathsf{a}}}\right], \left[\frac{q_{\mathsf{a}} 1}{q_{\mathsf{a}}}\right], \left[\frac{q_{\mathsf{a}} \sqcup}{q_{\mathsf{a}}}\right], \left[\frac{q_{\mathsf{a}} \#\#}{\#}\right]$$

The set of dominoes

- The set of dominoes

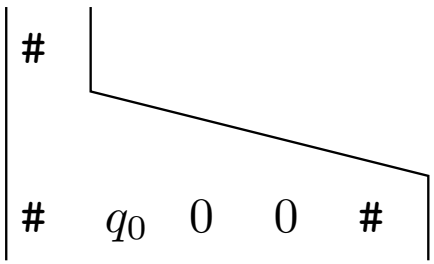
$$\left\{ \left[\frac{\#}{\#q_0 00\#}\right], \left[\frac{q_0 0}{0 q_1}\right], \left[\frac{q_0 1}{1 q_\mathsf{r}}\right], \left[\frac{q_1 0}{0 q_\mathsf{a}}\right], \left[\frac{q_0 \sqcup}{\sqcup q_\mathsf{r}}\right], \left[\frac{q_1 \sqcup}{\sqcup q_\mathsf{r}}\right], \left[\frac{0 q_1 1}{q_0 01}\right], \right.$$
$$\left[\frac{1 q_1 1}{q_0 11}\right], \left[\frac{\sqcup q_1 1}{q_0 \sqcup 1}\right], \left[\frac{0}{0}\right], \left[\frac{1}{1}\right], \left[\frac{\sqcup}{\sqcup}\right], \left[\frac{\#}{\#}\right], \left[\frac{\#}{\sqcup\#}\right], \left[\frac{0 q_\mathsf{a}}{q_\mathsf{a}}\right], \left[\frac{1 q_\mathsf{a}}{q_\mathsf{a}}\right],$$
$$\left. \left[\frac{\sqcup q_\mathsf{a}}{q_\mathsf{a}}\right], \left[\frac{q_\mathsf{a} 0}{q_\mathsf{a}}\right], \left[\frac{q_\mathsf{a} 1}{q_\mathsf{a}}\right], \left[\frac{q_\mathsf{a} \sqcup}{q_\mathsf{a}}\right], \left[\frac{q_\mathsf{a} \#\#}{\#}\right] \right\}$$

- It was purely constructed from the definition of TM M and input string 00
- Given any TM M and a string w, we can use the same process to construct a set of dominoes
- Let's play

The set of dominoes

- Let's focus on the set of dominoes that we just constructed:

$$\left\{ \left[\frac{\#}{\#q_0 00\#}\right], \left[\frac{q_0 0}{0 q_1}\right], \left[\frac{q_0 1}{1 q_r}\right], \left[\frac{q_1 0}{0 q_a}\right], \left[\frac{q_0 \sqcup}{\sqcup q_r}\right], \left[\frac{q_1 \sqcup}{\sqcup q_r}\right], \left[\frac{0 q_1 1}{q_0 01}\right], \right.$$
$$\left[\frac{1 q_1 1}{q_0 11}\right], \left[\frac{\sqcup q_1 1}{q_0 \sqcup 1}\right], \left[\frac{0}{0}\right], \left[\frac{1}{1}\right], \left[\frac{\sqcup}{\sqcup}\right], \left[\frac{\#}{\#}\right], \left[\frac{\#}{\sqcup\#}\right], \left[\frac{0 q_a}{q_a}\right], \left[\frac{1 q_a}{q_a}\right],$$
$$\left. \left[\frac{\sqcup q_a}{q_a}\right], \left[\frac{q_a 0}{q_a}\right], \left[\frac{q_a 1}{q_a}\right], \left[\frac{q_a \sqcup}{q_a}\right], \left[\frac{q_a \# \#}{\#}\right] \right\}$$



The set of dominoes

- Let's focus on the set of dominoes that we just constructed:

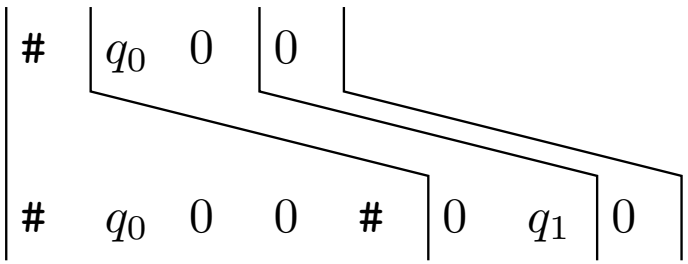
$$\left\{ \left[\frac{\#}{\#q_0 00\#}\right], \left[\frac{q_0 0}{0 q_1}\right], \left[\frac{q_0 1}{1 q_r}\right], \left[\frac{q_1 0}{0 q_a}\right], \left[\frac{q_0 \sqcup}{\sqcup q_r}\right], \left[\frac{q_1 \sqcup}{\sqcup q_r}\right], \left[\frac{0 q_1 1}{q_0 01}\right], \right.$$
$$\left[\frac{1 q_1 1}{q_0 11}\right], \left[\frac{\sqcup q_1 1}{q_0 \sqcup 1}\right], \left[\frac{0}{0}\right], \left[\frac{1}{1}\right], \left[\frac{\sqcup}{\sqcup}\right], \left[\frac{\#}{\#}\right], \left[\frac{\#}{\sqcup\#}\right], \left[\frac{0 q_a}{q_a}\right], \left[\frac{1 q_a}{q_a}\right],$$
$$\left. \left[\frac{\sqcup q_a}{q_a}\right], \left[\frac{q_a 0}{q_a}\right], \left[\frac{q_a 1}{q_a}\right], \left[\frac{q_a \sqcup}{q_a}\right], \left[\frac{q_a \# \#}{\#}\right] \right\}$$

#	q_0	0				
#	q_0	0	0	#	0	q_1

The set of dominoes

- Let's focus on the set of dominoes that we just constructed:

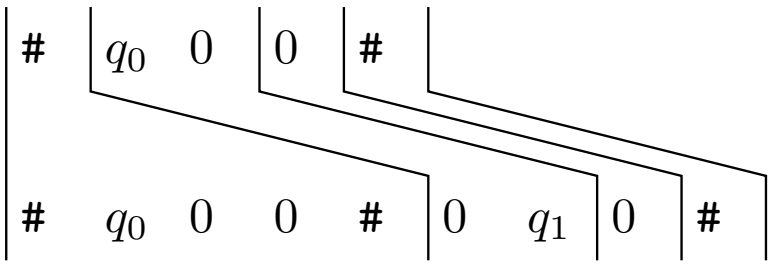
$$\left\{ \left[\frac{\#}{\#q_0 00\#}\right], \left[\frac{q_0 0}{0 q_1}\right], \left[\frac{q_0 1}{1 q_r}\right], \left[\frac{q_1 0}{0 q_a}\right], \left[\frac{q_0 \sqcup}{\sqcup q_r}\right], \left[\frac{q_1 \sqcup}{\sqcup q_r}\right], \left[\frac{0 q_1 1}{q_0 01}\right], \right.$$
$$\left[\frac{1 q_1 1}{q_0 11}\right], \left[\frac{\sqcup q_1 1}{q_0 \sqcup 1}\right], \left[\frac{0}{0}\right], \left[\frac{1}{1}\right], \left[\frac{\sqcup}{\sqcup}\right], \left[\frac{\#}{\#}\right], \left[\frac{\#}{\sqcup\#}\right], \left[\frac{0 q_a}{q_a}\right], \left[\frac{1 q_a}{q_a}\right],$$
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The set of dominoes

- Let's focus on the set of dominoes that we just constructed:

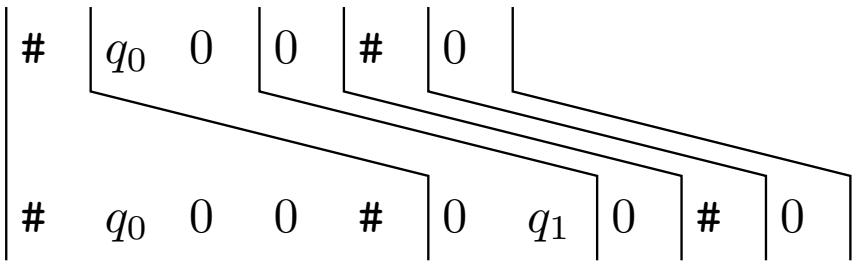
$$\left\{ \left[\frac{\#}{\#q_0 00\#}\right], \left[\frac{q_0 0}{0 q_1}\right], \left[\frac{q_0 1}{1 q_r}\right], \left[\frac{q_1 0}{0 q_a}\right], \left[\frac{q_0 \sqcup}{\sqcup q_r}\right], \left[\frac{q_1 \sqcup}{\sqcup q_r}\right], \left[\frac{0 q_1 1}{q_0 01}\right], \right.$$
$$\left[\frac{1 q_1 1}{q_0 11}\right], \left[\frac{\sqcup q_1 1}{q_0 \sqcup 1}\right], \left[\frac{0}{0}\right], \left[\frac{1}{1}\right], \left[\frac{\sqcup}{\sqcup}\right], \left[\frac{\#}{\#}\right], \left[\frac{\#}{\sqcup\#}\right], \left[\frac{0 q_a}{q_a}\right], \left[\frac{1 q_a}{q_a}\right],$$
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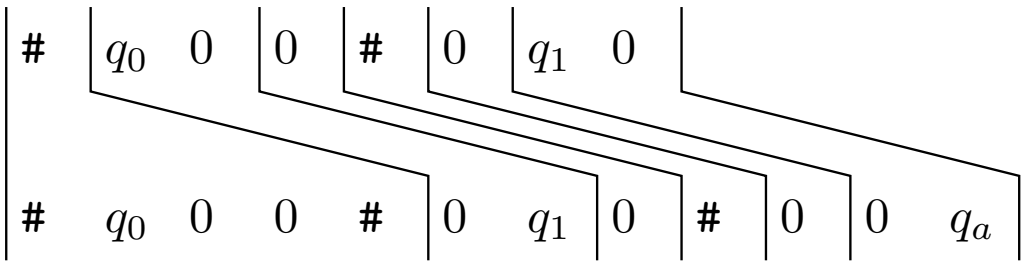
$$\left\{ \left[\frac{\#}{\#q_0 00\#}\right], \left[\frac{q_0 0}{0 q_1}\right], \left[\frac{q_0 1}{1 q_r}\right], \left[\frac{q_1 0}{0 q_a}\right], \left[\frac{q_0 \sqcup}{\sqcup q_r}\right], \left[\frac{q_1 \sqcup}{\sqcup q_r}\right], \left[\frac{0 q_1 1}{q_0 01}\right], \right.$$
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The set of dominoes

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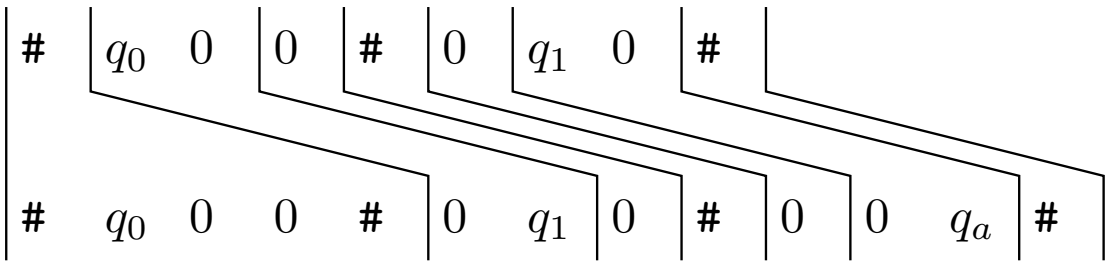
$$\left\{ \left[\frac{\#}{\#q_0 00\#}\right], \left[\frac{q_0 0}{0 q_1}\right], \left[\frac{q_0 1}{1 q_r}\right], \left[\frac{q_1 0}{0 q_a}\right], \left[\frac{q_0 \sqcup}{\sqcup q_r}\right], \left[\frac{q_1 \sqcup}{\sqcup q_r}\right], \left[\frac{0 q_1 1}{q_0 01}\right], \right.$$
$$\left[\frac{1 q_1 1}{q_0 11}\right], \left[\frac{\sqcup q_1 1}{q_0 \sqcup 1}\right], \left[\frac{0}{0}\right], \left[\frac{1}{1}\right], \left[\frac{\sqcup}{\sqcup}\right], \left[\frac{\#}{\#}\right], \left[\frac{\#}{\sqcup\#}\right], \left[\frac{0 q_a}{q_a}\right], \left[\frac{1 q_a}{q_a}\right],$$
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The set of dominoes

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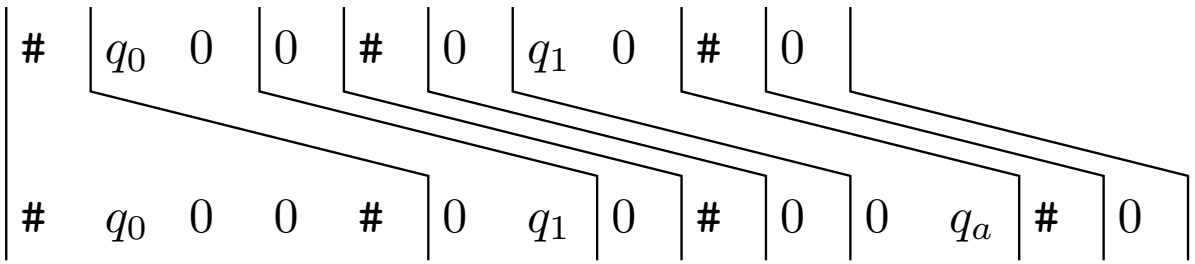
$$\left\{ \left[\frac{\#}{\#q_0 00\#}\right], \left[\frac{q_0 0}{0 q_1}\right], \left[\frac{q_0 1}{1 q_r}\right], \left[\frac{q_1 0}{0 q_a}\right], \left[\frac{q_0 \sqcup}{\sqcup q_r}\right], \left[\frac{q_1 \sqcup}{\sqcup q_r}\right], \left[\frac{0 q_1 1}{q_0 01}\right], \right.$$
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The set of dominoes

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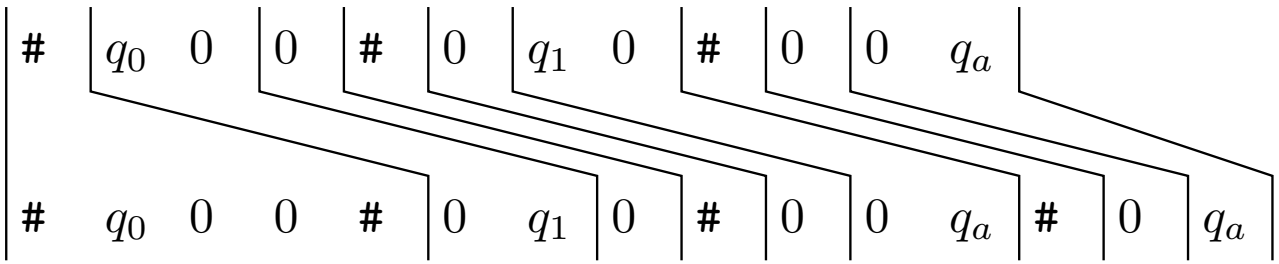
$$\left\{ \left[\frac{\#}{\#q_0 00\#}\right], \left[\frac{q_0 0}{0 q_1}\right], \left[\frac{q_0 1}{1 q_r}\right], \left[\frac{q_1 0}{0 q_a}\right], \left[\frac{q_0 \sqcup}{\sqcup q_r}\right], \left[\frac{q_1 \sqcup}{\sqcup q_r}\right], \left[\frac{0 q_1 1}{q_0 01}\right], \right.$$
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The set of dominoes

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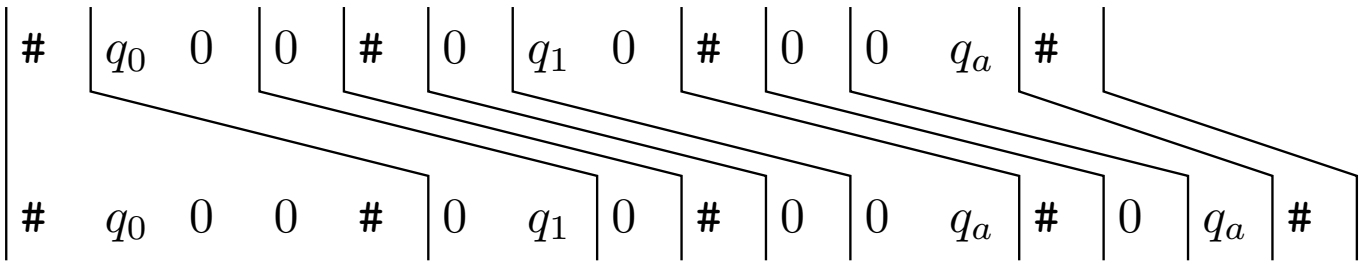
$$\left\{ \left[\frac{\#}{\#q_0 00\#}\right], \left[\frac{q_0 0}{0 q_1}\right], \left[\frac{q_0 1}{1 q_r}\right], \left[\frac{q_1 0}{0 q_a}\right], \left[\frac{q_0 \sqcup}{\sqcup q_r}\right], \left[\frac{q_1 \sqcup}{\sqcup q_r}\right], \left[\frac{0 q_1 1}{q_0 01}\right], \right.$$
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The set of dominoes

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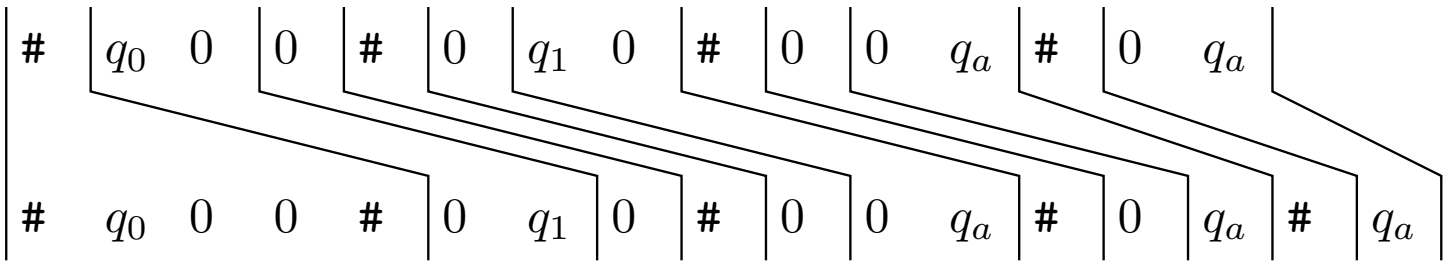
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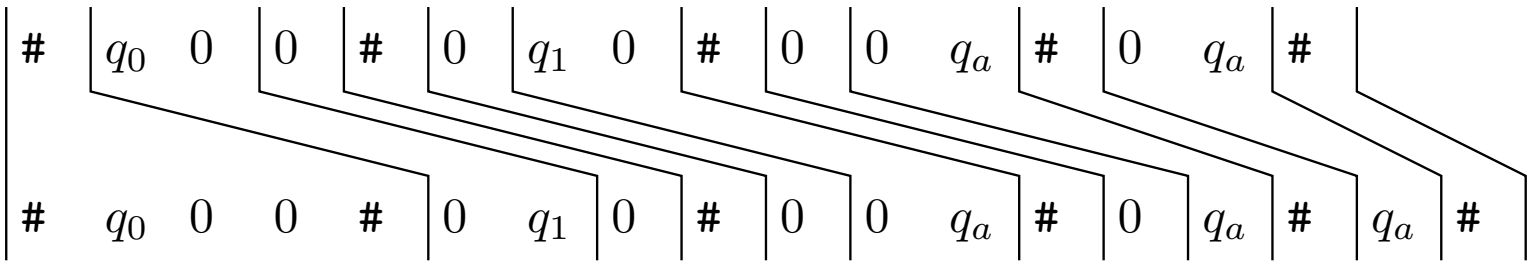
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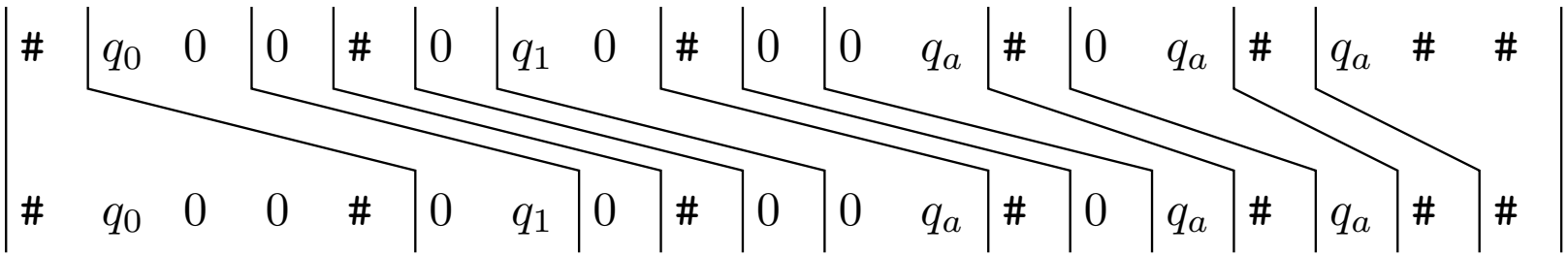
$$\left\{ \left[\frac{\#}{\#q_0 00\#}\right], \left[\frac{q_0 0}{0 q_1}\right], \left[\frac{q_0 1}{1 q_r}\right], \left[\frac{q_1 0}{0 q_a}\right], \left[\frac{q_0 \sqcup}{\sqcup q_r}\right], \left[\frac{q_1 \sqcup}{\sqcup q_r}\right], \left[\frac{0 q_1 1}{q_0 01}\right], \right.$$
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The set of dominoes

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The set of dominoes

- The set of dominoes

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- The above set of dominoes was constructed from TM M and the string 00
  - If M accepts 00, the accepting computation history for M on input 00 exists, results in a match
  - If M does not accept 00, the accepting computation history for M on input 00 does not exists, results in no match

The set of dominoes

- The set of dominoes

$$\left\{ \left[\frac{\#}{\#q_0 00\#}\right], \left[\frac{q_0 0}{0 q_1}\right], \left[\frac{q_0 1}{1 q_\mathsf{r}}\right], \left[\frac{q_1 0}{0 q_\mathsf{a}}\right], \left[\frac{q_0 \sqcup}{\sqcup q_\mathsf{r}}\right], \left[\frac{q_1 \sqcup}{\sqcup q_\mathsf{r}}\right], \left[\frac{0 q_1 1}{q_0 01}\right], \right.$$
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- Unfortunately, the above set of dominoes is an instance of *MPCP* not *PCP*
- If the above set of dominoes is an instance of *PCP* it has a match regardless of whether M accepts 00
  - Can you see one?

Converting P' of *MPCP* to P of *PCP*

- Now we have an instance P' of MPCP. We need to convert it to an instance of PCP.
- We need to enforce that the only match of P must start with a specific domino only.
- Let $u = u_1 u_2 \cdots u_n$ be any string of length n. Let

$$\begin{aligned} \star u &= * u_1 * u_2 * u_3 * \cdots * u_n \\ u\star &= u_1 * u_2 * u_3 * \cdots * u_n * \\ \star u\star &= * u_1 * u_2 * u_3 * \cdots * u_n * \end{aligned}$$

- Example: Suppose $u = 011$

$$\begin{aligned} \star u &= * 0 * 1 * 1 \\ u\star &= 0 * 1 * 1 * \\ \star u\star &= * 0 * 1 * 1 * \end{aligned}$$

Converting P' of *MPCP* to P of *PCP*

- If P' were the collection

$$\left\{\left[\frac{t_1}{b_1}\right], \left[\frac{t_2}{b_2}\right], \left[\frac{t_3}{b_3}\right], \ldots, \left[\frac{t_k}{b_k}\right]\right\}$$

  we let P be the collection

$$\left\{\left[\frac{\star t_1}{\star b_1 \star}\right], \left[\frac{\star t_1}{b_1 \star}\right], \left[\frac{\star t_2}{b_2 \star}\right], \left[\frac{\star t_3}{b_3 \star}\right], \ldots, \left[\frac{\star t_k}{b_k \star}\right], \left[\frac{\star \diamond}{\diamond}\right]\right\}$$

- In doing so, the match can only start with

$$\left[\frac{\star t_1}{\star b_1 \star}\right]$$

  The only domino that top and bottom start with the same symbol ($*$)

- The domino $\left[\frac{\star \diamond}{\diamond}\right]$ allows us to add extra $*$ to the top at the end of the match.

Conclusion

- Assume that *PCP* is decidable. Thus, there exists TM R decides *PCP*.
- Construct TM S that decides A_{TM}

  $S =$ "On input $\langle M, w \rangle$ where M is a TM and w is a string:
  1. Construct a set of dominoes P an instance of PCP from TM M and input w as we discussed
  2. Then run R on input $\langle P \rangle$
  3. If R accepts $\langle P \rangle$, *accept*
     If R rejects $\langle P \rangle$, *reject*
- To prove that TM S a decider for A_{TM} is straightforward

Conclusion

- Show that TM S is a decider for A_{TM}
  1. Assume that $\langle M, w\rangle \in A_{\mathsf{TM}}$. Since $\langle M, w\rangle \in A_{\mathsf{TM}}$, M accepts w. Since M accepts w, the set of dominoes P constructed in step 1 will have a match. In other words, $\langle P\rangle \in \mathit{PCP}$. Since R is a decider for *PCP*, by running R on input $\langle P\rangle$, R will accept $\langle P\rangle$. Since R accepts $\langle P\rangle$, S accepts $\langle M, w\rangle$.
  2. Assume that $\langle M, w\rangle \notin A_{\mathsf{TM}}$. Since $\langle M, w\rangle \notin A_{\mathsf{TM}}$, M does not accept w. Since M does not accept w, the set of dominoes P constructed in step 1 will have not have match. In other words, $\langle P\rangle \notin \mathit{PCP}$. Since R is a decider for *PCP*, by running R on input $\langle P\rangle$, R will reject $\langle P\rangle$. Since R rejects $\langle P\rangle$, S rejects $\langle M, w\rangle$.
- This shows that S is a decider for A_{TM}. Therefore, A_{TM} is decidable — contradiction
- *PCP* is undecidable